

Programme	B. Sc. Mathematics Honours			
Course Code	MAT8EJ406			
Course Title	OPERATIONS RESEARCH			
Type of Course	Major			
Semester	VIII			
Academic Level	400-499			
Course Details	Credit	Lecture/Tutorial per week	Practical per week	Total Hours
	4	4	-	60
Pre-requisites	Basic Mathematical and Statistical knowledge.			
Course Summary	This paper on Operation Research introduces the concepts like minimum path problem in network analysis, integer linear programming problem and dynamic programming problem. Kuhn Tucker condition to solve nonlinear programming problem is also discussed.			

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Solve Minimum Path Problem, Maximum flow problem	Ap	C	Internal Exam/ Assignment / Seminar/ Viva / End Sem Exam
CO2	Understand and solve ILP and MILP	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
CO3	Apply Kuhn-Tucker Conditions to solve nonlinear programming problem	Ap	P	Internal Exam/Assignment/ Seminar/ Viva / End Sem Exam
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Text: Optimization Methods in Operation Research and System Analysis (4th edition), KV Mittal, C Mohan, New Age International (P) Limited (2016)				
Module	Unit	Content	Hrs (48 +12)	Ext. Marks (70)
I		Flow and Potential in Networks	14	
	1	5.1,5.2 - Graphs Definitions and Notation		
	2	5.3- Minimum Path Problem		
	3	5.4- Spanning tree of minimum length		
	4	5.5- Problem of Potential Difference		
	5	5.6- Scheduling of sequential activities		
	6	5.7 Maximum flow problem		
	7	Generalized Problem of Maximum flow		
II		Integer Programming	10	
	8	6.1, 6.2-Introduction, ILP in two dimensional space		
	10	6.3-General ILP and MILP problems		
	11	6.4- Examples of ILP in two dimensional space		
	12	6.5,6.6, 6.7- Cutting planes, Example, Remarks on Cutting plane method		
III		Kuhn-Tucker Theory and Nonlinear Programming	11	
	14	8.1, 8.2-Introduction , Lagrangian Function: Saddle Point,		
	15	8.3- Relation between Saddle Point of $F(X,Y)$ and Minimal point of $f(X)$		
	16	8.4- Kuhn-Tucker Conditions		
	17	8.5- Primal and Dual Problems		
	18	8.6-Quadratic Programming		
IV		Dynamic Programming	13	
	19	10.1,10.2- Introduction, Problem 1: A Minimum Path Problem		

	20	10.3-Problem II: Single Additive Constraint, Additively Separable Return		
	21	10.4, 10.5-Problem III: Single Multiplicative Constraint, Additively Separable Return, Problem IV: Single Additive Constraint, Multiplicatively Separable Return		
	22	10.6,10.7-Computational Economy in DP, Serial Multistage Model		
	23	10.8, 10.9-Examples of Failure, Decomposition		
	24	10.10-Backward and Forward Recursion		
V	Open Ended		12	
	Sensitivity Analysis, Changes in b_i , c_j , and a_{ij} , Introduction of new variable, Introduction of new constraint, Deletion of variables, Deletion of constraints, Parametric linear programming, goal programming			
References:				
1. G. Hadley: Linear Programming Addison-Wesley Pub Co Reading, Mass (1975)				
2. G. Hadley : Non-linear and Dynamic Programming Wiley Eastern Pub Co. Reading, Mass (1964)				
3. S.S. Rao : Optimization – Theory and Applications (2nd Edn.) Wiley Eastern (P) Ltd. New Delhi.				
4. Russel L Ackoff and : Fundamentals of Operation Research Maurice W.Sasioni Wiley Eastern Ltd. New Delhi. (1991)				

Note: 70 external marks are distributed over the first four modules subjected to a minimum of 15 marks from each module

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	2	2	1	2	1	1	1	2	0	1
CO 2	3	3	1	1	2	1	1	1	2	0	1
CO 3	2	3	2	1	2	1	1	1	2	0	1

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Assignment/ Seminar
- Internal Exam
- Viva
- Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓